

SALESPERSON PERFORMANCE ANALYSIS

Task 27 | Veda Technology Internship | Data Analytics Track (Level 2, Day 27)

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TOTAL SALES	TOTAL PROFIT	MARGIN	ORDERS	AVG ORDER	#1 REP	LAST RANK
\$6.56M	\$461K	7.0%	3,047	\$2,152	Divya	Sameer

1. Objective and Approach

Task: Compare salespeople using sales, profit, growth and average order value. **Objective:** evaluate performance fairly. **Deliverables:** ranking, dashboard and recommendations.

Ranking salespeople on total sales alone is unfair, because a rep in a large or high-margin territory looks better than a rep in a tougher one. This analysis therefore uses **multiple KPIs** and adjusts for **territory differences**, then combines them into one Fair Score (0 to 100).

Fair Score weights

Total Sales	Total Profit	Profit Margin	Growth (CAGR 14-17)	Avg Order Value	Share of Region Sales	Margin vs Region
15%	20%	15%	15%	10%	10%	15%

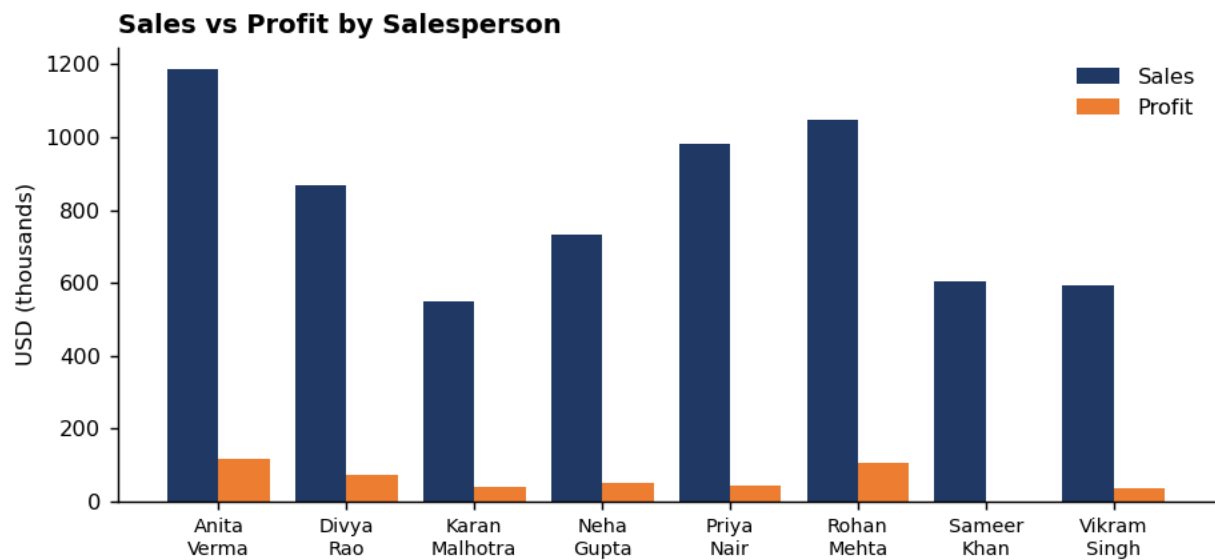
Each KPI is first scaled between 0 (lowest rep) and 1 (highest rep), then multiplied by its weight. Growth uses the 3-year CAGR (2014-2017) because one year-on-year number is too volatile. Share of Region Sales and Margin vs Region compare each rep with the other rep in the same territory (two reps per region).

2. Executive Summary

- **Divya Rao** ranks #1 (score 75.3), narrowly ahead of **Anita Verma** (75.1). Divya Rao would be only #4 on sales alone, but leads on growth (25.1% CAGR) and average order value (\$2,814).
- **Sameer Khan** ranks last (score 22.6). Sales of \$604,716 produced only \$460 profit (margin 0.1%), even though growth is strong (24.1% CAGR).
- **Discounts destroy profit:** 1,315 line items with 30% or more discount made -\$192,840 in total, while items with 20% or less discount earned \$654,316.
- **Territory matters:** West margin is 9.9% but Central is only 3.7%. Reps in weaker regions are judged against their own territory.
- **Profitability is slipping:** company sales rose from \$1.49M (2014) to \$1.72M (2017), but margin fell from 7.7% to 6.0%.

3. Salesperson KPI Comparison

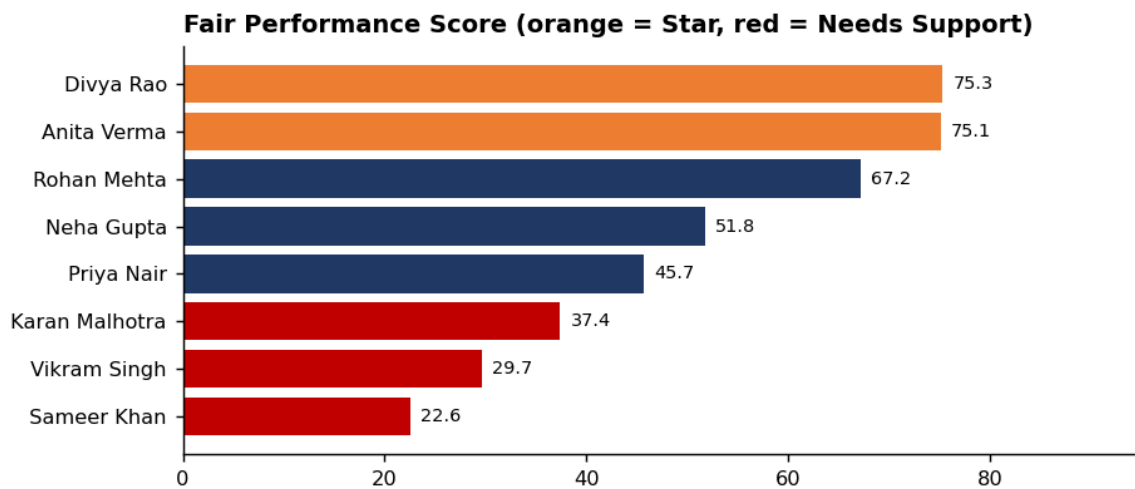
Salesperson	Region	Sales	Profit	Margin	Orders	Avg Order	Avg Disc.	YoY 17 vs 16	CAGR 14-17
Anita Verma	West	\$1,186,289	\$116,108	9.8%	459	\$2,585	15.7%	-19.9%	1.6%
Divya Rao	South	\$866,636	\$71,307	8.2%	308	\$2,814	15.3%	32.8%	25.1%
Karan Malhotra	East	\$547,353	\$40,678	7.4%	416	\$1,316	15.5%	10.2%	8.3%
Neha Gupta	Central	\$731,119	\$48,930	6.7%	340	\$2,150	16.5%	61.5%	-0.9%
Priya Nair	East	\$980,147	\$41,276	4.2%	422	\$2,323	16.3%	-18.6%	-4.8%
Rohan Mehta	West	\$1,048,633	\$105,960	10.1%	490	\$2,140	17.5%	-0.2%	3.5%
Sameer Khan	Central	\$604,716	\$460	0.1%	332	\$1,821	17.1%	-11.7%	24.1%
Vikram Singh	South	\$592,218	\$36,757	6.2%	280	\$2,115	15.9%	18.0%	-3.5%
Company		\$6,557,110	\$461,476	7.0%	3,047	\$2,152		2.8%	4.9%



Anita Verma and Rohan Mehta lead on total sales and profit, but margins are similar across the top reps (about 10%). The big differences are in growth, order value and the Central and East reps with thin margins.

4. Final Ranking (Fair Score)

Rank	Salesperson	Region	Fair Score	Sales-only Rank	Rank Change	Share of Region Sales	Margin vs Region	Tier
1	Divya Rao	South	75.3	4	+3	59.4%	+0.8%	Star Performer
2	Anita Verma	West	75.1	1	-1	53.1%	-0.1%	Star Performer
3	Rohan Mehta	West	67.2	2	-1	46.9%	+0.2%	Solid Contributor
4	Neha Gupta	Central	51.8	5	+1	54.7%	+3.0%	Solid Contributor
5	Priya Nair	East	45.7	3	-2	64.2%	-1.2%	Solid Contributor
6	Karan Malhotra	East	37.4	8	+2	35.8%	+2.1%	Needs Support
7	Vikram Singh	South	29.7	7	=	40.6%	-1.2%	Needs Support
8	Sameer Khan	Central	22.6	6	-2	45.3%	-3.6%	Needs Support

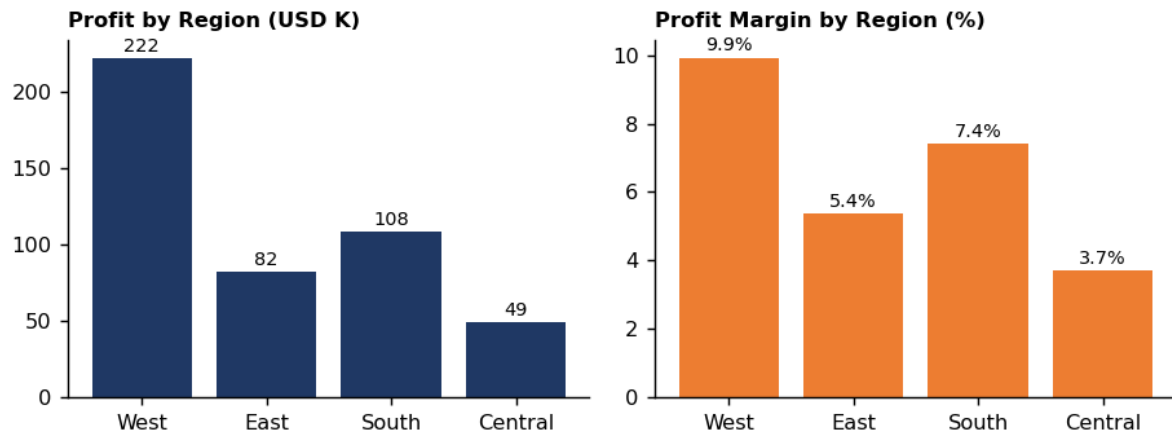


How to read this

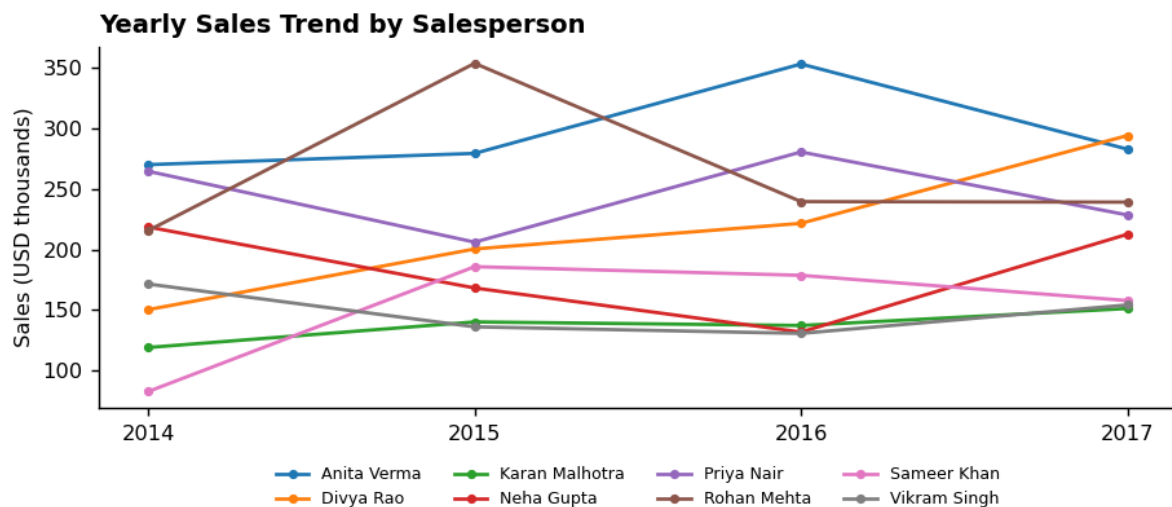
- **Rank Change** shows how the rank moves compared with a sales-only ranking. A plus sign means the rep is better than sales alone suggests.
- **Divya Rao** moves from #4 on sales to #1: fast growth (25.1% CAGR), the highest average order value (\$2,814) and 59.4% of her region's sales.
- **Priya Nair** falls from #3 to #5 because her margin (4.2%) is low and sales fell 18.6% in 2017.
- The two Star Performers are close (75.3 vs 75.1), so they should be treated as joint leaders rather than a clear winner and runner-up.

5. Territory and Yearly Trend

Region	Sales	Profit	Margin
West	\$2,234,922	\$222,068	9.9%
East	\$1,527,500	\$81,955	5.4%
South	\$1,458,853	\$108,064	7.4%
Central	\$1,335,835	\$49,390	3.7%



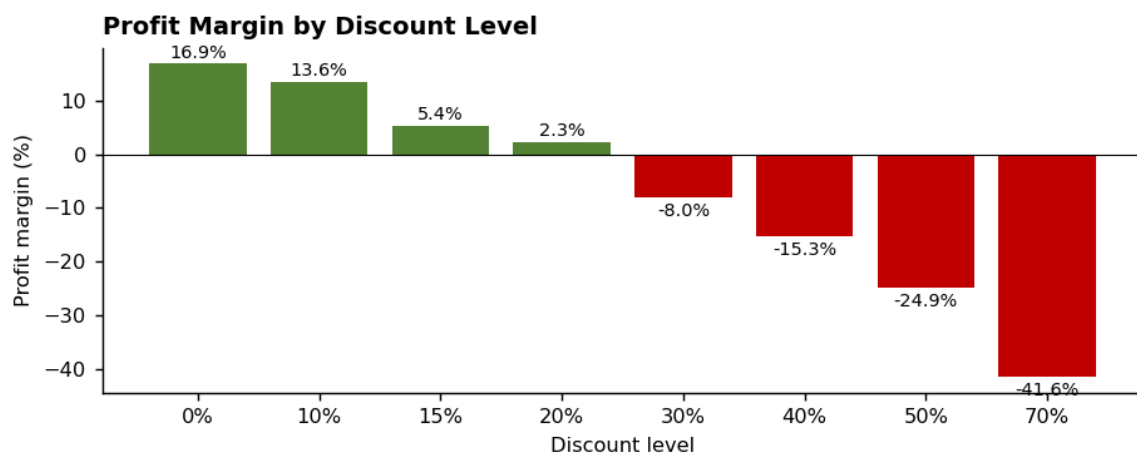
West earns 9.9% margin on \$2.23M sales while Central earns 3.7%. Neha Gupta (Central) beats her region margin by +3.0 points and Sameer Khan trails it by -3.6 points, so judging both against the company average would be unfair.



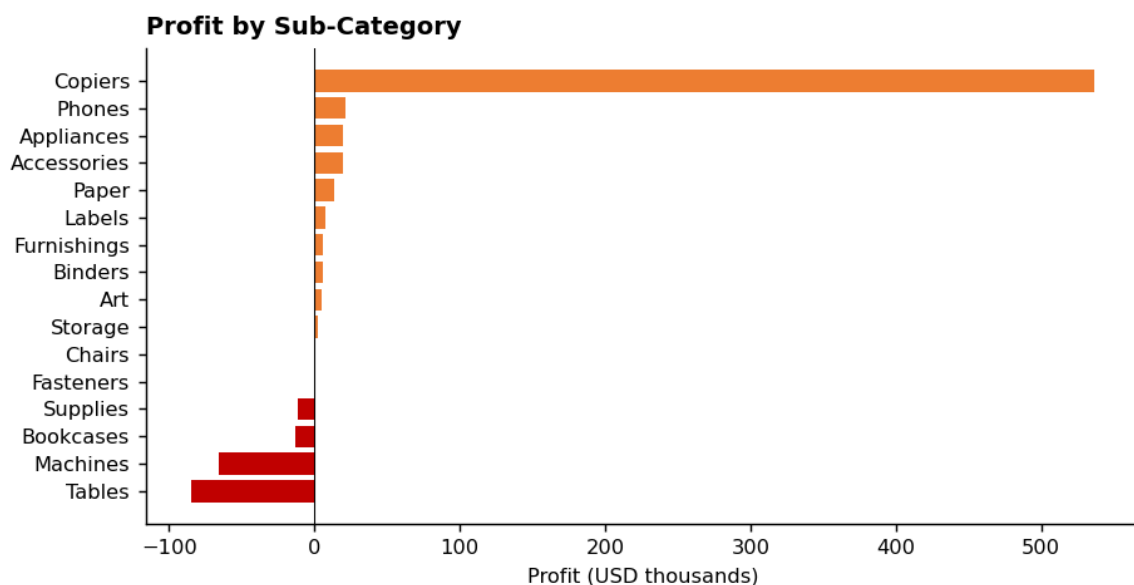
Year	Sales	Profit	Margin
2014	\$1,492,442	\$115,033	7.7%
2015	\$1,670,284	\$126,682	7.6%
2016	\$1,673,580	\$116,568	7.0%
2017	\$1,720,804	\$103,192	6.0%

6. Discount and Product Analysis

Discount	Line Items	Sales	Profit	Margin
0%	2,325	\$3,254,279	\$549,987	16.9%
10%	269	\$404,633	\$54,933	13.6%
15%	241	\$315,406	\$16,970	5.4%
20%	1,051	\$1,383,002	\$32,426	2.3%
30%	414	\$498,487	-\$40,013	-8.0%
40%	428	\$401,410	-\$61,596	-15.3%
50%	267	\$200,248	-\$49,789	-24.9%
70%	206	\$99,644	-\$41,443	-41.6%



Profit turns negative at 30% discount. Items sold at 30% or more lost \$192,840 in total (margin -16.1%), which wipes out a large part of the \$654,316 earned on items at 20% or less.



Loss-making sub-categories: **Tables** (-\$85,052), **Machines** (-\$65,938), **Bookcases** (-\$13,629), **Supplies** (-\$11,202). Technology is 71.6% of sales and Furniture has a negative margin (-7.4%).

7. Recommendations

#	Area	Recommendation
1	Reward on the Fair Score, not on sales alone	Recognise Divya Rao and Anita Verma as joint top performers and use them as peer mentors. Link bonuses to the Fair Score so reps in tough territories are not penalised.
2	Cap discounts at 20%	Require manager approval above 20% and stop deals above 30% unless the stock is being cleared. This is the single biggest profit lever in the data.
3	Fix loss-making products	Reprice, renegotiate supplier cost or limit discounts on Tables, Machines, Bookcases, Supplies.
4	Coach the weak territory (Central)	Set region-specific margin targets. Review Sameer Khan: strong growth (24.1% CAGR) but only \$460 profit on \$604,716 sales, which points to heavy discounting.
5	Invest in high-growth talent	Give Divya Rao larger accounts and support. She has the highest 3-year growth and the highest average order value.
6	Investigate declining sales	Anita Verma (-19.9%) and Priya Nair (-18.6%) both dropped sharply in 2017. Anita is the largest seller, so check for lost accounts first.
7	Raise order value	Train Karan Malhotra on bundling and upselling. His average order value (\$1,316) is far below the company average (\$2,152).
8	Reduce category concentration	Technology drives 71.6% of sales. Diversify the portfolio and track Machines margin separately.
9	Track margin every month	Sales are growing but margin fell from 7.7% to 6.0%. Add profit margin and discount level to the monthly sales review.

8. Data, Method and Limitations

- **Data:** a synthetic Superstore-style dataset (5,201 line items, 3,047 orders, 2014-2017, 22 columns) with an added Salesperson column. Figures are for practice and are not real company results.
- **Method:** all KPIs were calculated in Excel with SUMIFS, COUNTIFS and RANK formulas. Orders are counted once using an Order Flag helper column. Weights are editable in the workbook.
- **Limitations:** two reps per region means the territory adjustment compares each rep with one peer only. Weights are a judgement call, and a different weighting can change the order of close ranks (for example #1 and #2).
- **Files:** Task27_Salesperson_Performance_Analysis.xlsx contains the live dashboard, ranking, KPI sheets and this analysis.